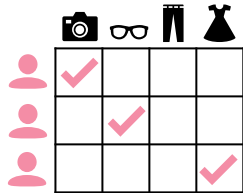


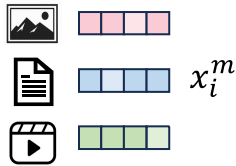
## Input

### u-i Interactions $R$



|  |   |   |  |   |
|--|---|---|--|---|
|  |   |   |  |   |
|  | ✓ |   |  |   |
|  |   | ✓ |  |   |
|  |   |   |  | ✓ |

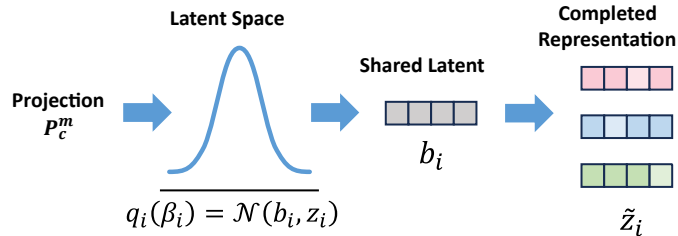
### Item Modalities



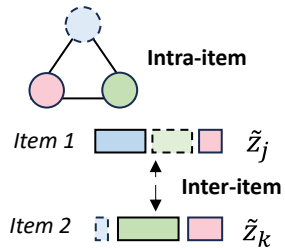
### User Embedding



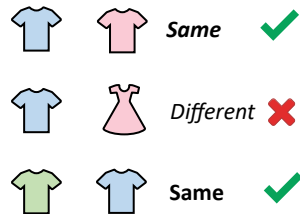
## Structural Imputation for Calibration (SIC)



### Structural Calibration

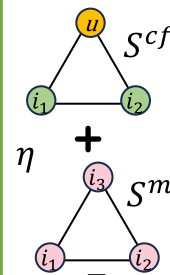


### Correspondence Calibration

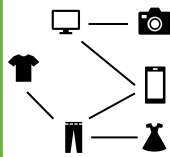


## Preference-oriented Representation Calibration (PRC)

### Relation Level

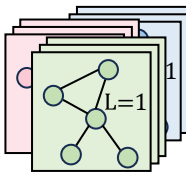


### Item-item Graph

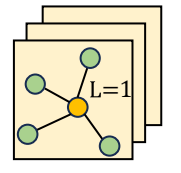


### GCN Propagation

#### Item-item Graph Propagation



#### User-item Graph Propagation



### Preference Level Alignment

#### Observed (real branch)



Align

#### Recovered (pseudo branch)



Same  $(i, m)$

## Recommendation Score

